STAT/MA 41600

In-Class Problem Set #34: November 6, 2026

1. Consider a collection of students who answer a survey about the holidays. Let X denote the percentage of students who plan to travel during the holidays. If X is a Beta random variable with $\alpha = 5$ and $\beta = 2$, find $P(X \geq 3/5)$. In other words, find the probability that at least 60% of the students plan to travel during the holidays.

2. We know $E(X) = \alpha/(\alpha + \beta)$, but it is "good practice" to derive it in specific cases, without using this formula. Use the probability density function of X from Question 1 (with $\alpha = 5$ and $\beta = 2$) to derive the value of $E(X)$ by integrating, i.e., by calculating: $E(X) = \int_0^1 x\, f_X(x)\, dx$.

3. Again use the assumptions from Question 1 for random variable X. Also consider a continuous uniform random variable U which is uniformly distributed on $(0, 1)$ and is independent of X. Calculate the probability that X is bigger than U, i.e., find the value of $P(X > U)$.

4. Consider a city in which the percentage of rainy days per year is modeled by a Beta random variable X with $\alpha = 3$ and $\beta = 2$. Find the probability that, during a given year, at least half of the days in this city are rainy.

5. Again use the assumptions from the previous question (i.e., Question 4) for random variable X. Calculate $P(|X - 1/2| < 2/5)$.